

Model-level architecture

One data interface feeds two backbone branches with different supervision paths

1 Inputs



image



document



chart



question +
choices

2 Shared model interface

Prompt template

Builds answer-only or rationale-plus-answer target text.

Multimodal processor

Encodes image placeholders and text with the same processor used by the model.

Span tags

Marks answer and rationale tokens so losses hit the correct target positions.

3A Qwen2-VL branch



Qwen2-VL + QLoRA

The model is loaded in 4-bit precision. Rank-16 QLoRA adapters update a small trainable part of the language-side projection and transformer stack.

Vision input

Image tokens are provided through the model processor.

Adapter updates

QLoRA changes only the adapter parameters, not the full backbone.

Generated text

Produces answer text, or rationale followed by answer.

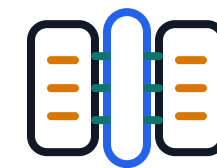
answer-only CE

rationale-generative CE

explanation-aware span CE

length-aware alpha control

3B BLIP-2 branch



BLIP-2 + Q-Former

The image encoder and OPT language model stay frozen. The Q-Former path and projection heads provide the trainable alignment point.

Frozen image encoder

Extracts visual features without full vision fine-tuning.

Q-Former features

Pooled query tokens become the image representation.

Answer projection

Answer-token features are projected for image-answer contrastive scoring.

generative CE

InfoNCE image-answer loss

retrieval diagnostic

frozen / generative / contrastive controls

4 Supervision signals



Target-token CE

Loss is applied to target tokens, not to the prompt.



Explanation-aware CE

Answer and rationale spans are weighted separately by alpha.



Contrastive-enhanced

BLIP-2 adds image-answer InfoNCE on top of generation.

5 Outputs and diagnostics



Answer score

Native metric for each dataset.



Generated rationale

Saved when the prompt and objective ask for one.



Evidence sensitivity

Masking drift checks whether answer scores change.